

Form and color visual perception in green exercise: Positive effects on attention, mood, and self-esteem

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ABSTRACT

Physical activity performed in a natural environment, especially among green spaces, is associated with mental health benefits. Yet few studies have explored the separable influences of visual perception on the effects of green exercise. The present study assessed the extent to which two components of visual perception (form and color) contributed to mental health benefits by separating the form and color characteristics of an image of a green tree. In total, 76 healthy college students (18–24 years of age) engaged in incremental cycling exercise at a moderate intensity while viewing an image in front of them on a large screen. Participants were randomly assigned to one of four groups in which the image they viewed was a green tree, an achromatic tree, a formless color green, or a formless achromatic color. Before and after cycling (test time), participants' selective attention, mood state, and self-esteem were assessed using the Perception and Attention Function battery and the Chinese versions of the Profile of Mood States scale and the Rosenberg Self-Esteem Scale, respectively. Significant interactions detected between test time and form perception indicated that scores in total mood disturbance decreased and scores in the esteem subscale and self-esteem scale increased after cycling while viewing trees. A significant interaction between test time and color perception indicated that anger decreased in the two green color conditions. A significant interaction of color perception by form perception by test time was associated with a significant decrease in response times on the attention task after cycling among participants who viewed the green tree. Our findings are important for interpreting mechanisms underlying characteristics attributable to environmental benefits and revealed that single (color or form) or combined (color and form) visual perception improved the mental health benefits of exercise. These findings may also inform the design and planning of urban green spaces to maximize green exercise effects.

1. Introduction

Despite an increase in natural environment space or urban green spaces (i.e., community space comprising land, such as parks, rather than buildings) to promote residents' physical activity and public mental health (Byrka & Ryczko, 2018; Das & Gailey, 2022; Tyrväinen et al., 2014), few studies have investigated the mechanisms underlying the specific beneficial characteristics of green exercise. For health purposes, increasingly more urban green spaces are serving as natural environments within which physical activities can take place. Physical

activities, such as walking, running, and cycling, in natural environments (i.e., green exercise) have more synergistic benefits than those from exposure to other environments or from engaging in physical activity alone or being in a natural environment alone (Pretty et al., 2005). Green exercise has been widely shown to provide additional benefits to human mental health, including reducing stress (Rogerson et al., 2016) and mood disturbances (Gidlow et al., 2016; Pretty et al., 2007), improving self-esteem (Barton & Pretty, 2010; Pretty et al., 2005), affect (Das & Gailey, 2022), well-being (Rogerson et al., 2020) and enjoyment (Lahart et al., 2019), and restoring attention (Fraser et al., 2019).

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Landscape planners, designers, and researchers, therefore, should cultivate urban green space for physical activities, to promote physiological and psychological benefits.

Most empirical studies assessing the benefits of outdoor physical activity have focused on the availability, number, or presence of parks and green space (Garrett et al., 2020; Rioux et al., 2016). However, green exercise contributes to greater psychological benefits than exercise in other environments (Barton & Pretty, 2010). Psychological restoration theories, such as the Attention Restoration Theory (ART) and the Stress Reduction Theory (SRT), suggest that compared with non-green exercise, mental health benefits from green exercise are more restorative to public mental health. The ART suggests that restoration from directed attention fatigue contributes to the beneficial effects of green exercise and enables better cognitive function (Jones et al., 2021; R. Kaplan & Kaplan, 1989). The SRT suggests that exposure to nature can reduce stress, arousal, and negative emotions as well as promote recovery through psycho-physiological pathways (Ulrich, 1986; Ulrich et al., 1991). More specifically, people report more positive affect, less negative affect, and improved cognition after exposure to nature (Meidenbauer et al., 2020). In contrast, exposure to urban environments may produce anger and aggressiveness (Hartig et al., 2003). Thus, the restoration of natural environments may extend the mental health benefits of physical activity.

Although exposure to nature has been consistently associated with better mental health, evidence assessing the contributions of the specific characteristics of green exercise that are associated with benefits are scarce. For example, the relationship between street greenery and various health outcomes has been demonstrated (Lu, 2019). This raises an important issue of how to design green spaces in crowded urban areas to meet the demands of physical activity and maximize mental health benefits. In order to better address this issue, it is important to identify which natural attributes confer mental health benefits and to what extent perceptions account for these benefits. However, the specific natural attributes that provide mental health benefits remain unclear.

It has been hypothesized that some sensory perceptions of natural attributes are more important or preferred than others with respect to restoration of mental health (Grahn & Stigsdottir, 2010). Existing evidence has shown that visual perception is incredibly important during exposure to natural environments. Studies exploring the relationship between mental health benefits and visual perception often used images (Pretty et al., 2005) or videos (Rogerson & Barton, 2015). Compared with virtual urban or manufactured environments, visual exposure to real-life nature while participating in physical activities provides more mental health benefits, for example, recovery from stress (Brown et al., 2013) and improved affect (Olafsdottir et al., 2020), self-esteem (Brown et al., 2013), and directed attention (Rogerson & Barton, 2015). By contrast, clear effects on mental health are not observed following the same physical activities performed indoors (Rogerson & Barton, 2015). Such studies have provided preliminary evidence for the beneficial effects of green exercise while visually exposed to nature as well as the significance of visual perception on mental health benefits.

In fact, visual exposure to various natural environments does not always offer the same degree of restorative effects but appears to depend on the natural elements included (Celikors & Wells, 2022; Gatersleben & Andrews, 2013; White et al., 2010). For example, people respond positively to trees (Donovan et al., 2013; Nordh et al., 2011). The presence of trees has been associated with positive mental health, such as recovery (Ulrich, 1984), mood (Chi et al., 2022), well-being (Leather et al., 1998), and lower psychological distress (Zhang et al., 2022). Trees also encourage people to engage in more physical activity (Richardson et al., 2013). These findings suggest that trees may be an important natural element for physical activity and mental health benefits.

Although previous qualitative studies have explored the main traits (e.g., size, shade, color, and leafiness) associated with people's preferred trees (Camacho-Cervantes et al., 2014), there is a need for research that explores how the visual perception of trees in natural environments

contributes to the beneficial effects of green exercise. Akers et al. (2012) asked participants to perform moderate-intensity cycling tasks while watching video footage with a high percentage of foliage. The video footage was achromatic-filtered (gray), red-filtered (red), or unedited (green). The results showed lower mood disturbances and perceived exertion with viewing green foliage compared with viewing gray and red foliage. That study provided preliminary evidence that the color green itself may have a contributory effect to the benefits of green exercise. However, there is more than one visual feature in natural environments, namely, color and form. Thus, improvements in mood states may not critically depend on the color green. Indeed, this has been demonstrated in subsequent studies that failed to find positive effects of perceiving green on mood states (Briki & Majed, 2019). Thus, how visual features affect mental health warrants further investigation.

Therefore, the extent to which two components of visual perception—form and color—contribute to the effects of green exercise was the focus of the present study. Given that green exercise has a wide range of benefits, the present study assessed attention, mood state, and self-esteem as indicators of mental health. As constructivists assumed, the mental health promoting effects of nature are acquired through the semantic idea of nature and its restorative abilities. The recognition and association of features with the concept of nature may influence the preference and restoration of a specific natural environment (Van Hedger et al., 2019). Rich visual features that are more easily associated with the concept of nature may elicit more benefits. Therefore, we hypothesized that cycling while watching images of either the color green or the form of a tree would improve attention, mood state, and self-esteem. The findings of the present study will contribute to a better understanding of the key components of visual perception in green exercise.

2. Materials and methods

2.1. Design

We used a 2×2 between-group experimental study with pretest-posttest design to evaluate the effects of form and color perception on attention, mood state, and self-esteem. The independent variables were color perception (green vs. achromatic) and form perception (tree vs. blank). The dependent variables were attention, mood state, and self-esteem, data for which were collected before and after an incremental cycling exercise performed by participants while they viewed images of a green tree, an achromatic tree, a formless green color, or a formless achromatic color.

2.2. Participants

This study was approved by the Institutional Review Board at Tianjin University of Sport, and was conducted in accordance with the Declaration of Helsinki. All participants signed the informed consent, including experimental protocol, benefits, and harms of the study, before the experiment.

Using G*Power 3.1 software (Faul et al., 2007), an a priori power analysis for repeated-measures analysis of variance (ANOVA) within-between interaction, with a power of 0.80, an effect size f of 0.25, and a statistical significance level of 0.05, we found that a total sample of 48 participants was sufficient. These parameters were set with reference to previous studies (Brancato et al., 2022; Flowers et al., 2018). To account for potential drop out or participant data deletion, 76 healthy college students (40 males; 18–24 years of age) with normal color vision and normal or corrected-to-normal vision were recruited through an advertisement at the university campus. All participants had low risk of health issues when engaging in physical activity as assessed by the Physical Activity Risk Questionnaire (Thomas et al., 1992).

Participants were randomly assigned to one of four groups in a 1:1:1:1 ratio: (1) green tree (10 males and 9 females); (2) achromatic

tree (10 males and 9 females); (3) green color (10 males and 9 females); and (4) achromatic color (10 males and 9 females). No participants withdrew from the study. All participants successfully completed the evaluation and training sessions as allocated.

2.3. Experimental protocol

The experimental procedures are presented in Fig. 1. All participants were asked to refrain from caffeine, alcohol, and vigorous physical activity for at least 24 h and avoid eating for at least 3 h before measurements. All measurements were completed in a quiet room without distractions. Before the cycling period, participants performed baseline tests designed to measure their attention, mood state, and self-esteem. The order in which the tests were conducted was randomized for each participant. During the cycling period (details below), participants were given verbal instructions to pay attention to images of the green tree, achromatic tree, green color, or achromatic color projected onto a large screen (3.0×2.4 m) in front of them (2.5 m from the screen). After the cycling period, participants were asked to perform the same tests to measure their attention, mood state, and self-esteem in the same order as for the assessment of their baseline measurements.

2.4. Scenes

Video footage taken from real outdoor scenes best display the visual information of an environment. However, because such video footage includes several visual features (e.g., form and color), making it challenging to separate visual perception components, we altered these images to simulate natural settings but to present a single visual feature of the green element. We selected from the internet one image of a single green tree with a wide canopy and a short trunk because a study by Sommer and Summit (1995) demonstrated that people expressed preferences for trees with wider canopies and shorter trunks (Sommer & Summit, 1995). The image was unedited (green leaves) in the green tree group but was edited to be achromatic in the achromatic tree group. For the images shown to the green color and achromatic color groups, the entire screen was colored green (RGB: 0, 255, 0) or white (RGB: 255, 255, 255). All images were filtered or colored through Adobe Photoshop CS6 (Adobe Photoshop CS6, Adobe Systems, Inc.).

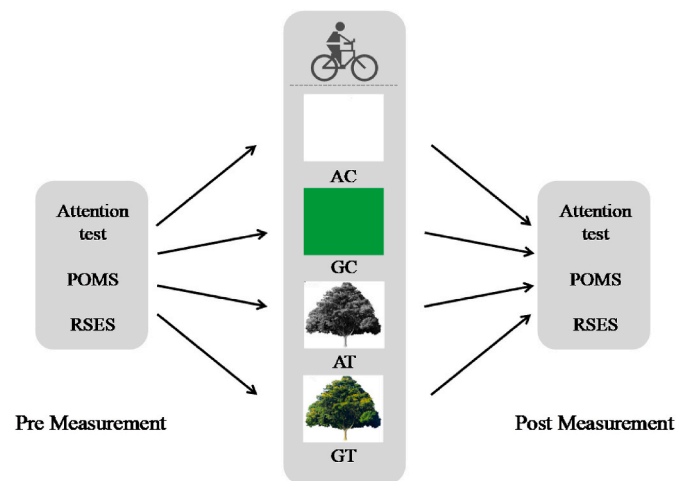


Fig. 1. Experimental procedure. Participants cycled at moderate intensity while viewing images on a screen of a formless achromatic color (AC), an achromatic tree (AT), a formless green color (GC), and a green tree (GT). POMS indicates Profile of Mood States Questionnaire, and RSES, Rosenberg Self-Esteem Scale. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

2.5. Cycling

Participants wrapped a Polar heart rate monitor (RS400, Polar Electro, Finland) around their chest and performed incremental cycling on a cycle ergometer (Ergoselect 100K, Ergoline, Germany). The exercise began at an intensity of 25 W, increased at a rate of 25 W every 2 min until perceived by the participant as “fairly light” on the Borg Rating of Perceived Exertion scale (Borg, 1998), and then cycled at this power for 20 min. During the cycling period, participants were asked to maintain a pedaling cadence of 70 rpm. The exercise intensity was monitored assisted by heart rate and perceived exertion in the whole cycling period to guarantee the moderate intensity (120–140 bpm) and avoid accidental injuries, which were not used for further data analysis.

2.6. Measurements

2.6.1. Attention

The Perception and Attention Function battery (designed to capture the subfunctions of attention) was used with the Vienna Test System (Schuhfried GmbH, Mödling, Austria) to assess selective attention. One geometric stimulus (circle, square, or triangle) that may get lighter or darker or stay the same was presented on a computer screen. Participants were required to press a key on the response panel as quickly as possible when circles and squares got lighter or darker, not in response to the other stimuli. Each stimulus was presented for 1500 ms, and a change might occur after 500 ms. The interval between stimuli was 1000 ms. The test consisted of 144 stimuli and lasted approximately 7 min. Participant’s response times (RTs) to the stimuli were recorded to assess selective attention.

2.6.2. Mood state

The Chinese version of the Profile of Mood States (POMS), a reliable and validated measurement tool, was used to assess mood states. Specifically, the POMS measured tension, anger, fatigue, depression, vigor, confusion, and esteem. Participants indicated how they felt “right now” about each item. The POMS scale contained 40 items, each with a 5-point Likert scale ranging from 0 (not at all) to 4 (extremely). POMS raw scores for each subscale were recorded as the sum of all items and converted into T-scores following the Chinese norm. The POMS total mood disturbance (TMD) score was calculated as follows: $TMD = [(tension + anger + fatigue + depression + confusion) - (vigor + esteem)] + 100$, with higher scores indicating more negative moods. Cronbach’s alpha was 0.66–0.83, indicating acceptable reliability (Zhu, 1995).

2.6.3. Self-esteem

The Chinese version of the Rosenberg Self-Esteem Scale (RSES) was used to assess self-esteem. The RSES consisted of 10 items that were each scored from 1 (strongly disagree) to 4 (strongly agree) points. Higher scores on the total scale indicate higher self-esteem. The Cronbach’s alpha was 0.82, indicating reasonably high reliability (Shen & Cai, 2008).

2.7. Statistical analysis

SPSS 27.0 statistics software (IBM Inc., Chicago, IL, USA) was used for all statistical analyses. Values are expressed as the mean $\pm$ standard deviation (SD) in tables and figures. The Shapiro-Wilk test was used to assess the normality of the distribution for all data sets. For non-normally distributed data sets, such as scores on the RSES and TMD and for the tension, anger, fatigue, depression, vigor, and confusion subscales, non-parametric tests were used. Kruskal-Wallis H tests were used to compare baseline measurements among groups. Generalized estimating equations (GEEs) were used to analyze the effects of test time (before vs. after cycling), form perception (blank vs. tree) and color perception (achromatic vs. green). For normally distributed data sets,

including scores on the esteem subscale and RTs on the attention test, one-way analyses of variance (ANOVAs) were used to evaluate the potential differences among groups in baseline measurements. We used $2 \times 2 \times 2$ repeated-measures ANOVAs to assess the main effects and interactions of test time (before vs. after cycling), form perception (blank vs. tree), and color perception (achromatic vs. green). *T*-tests were used for post-hoc analysis, as appropriate. The level of statistical significance for all tests was set at $p < .05$. Partial eta squared (η_p^2) and Cohen's *d* were calculated as effect size estimates.

3. Results

3.1. Baseline measurements

As shown in Table 1, there were no significant differences between groups for any baseline measurement (all *p* values $> .05$), suggesting that any subsequent differences between groups were likely attributable to different form and color perceptions.

3.2. Effects of form perception and color perception on measurements

All measurements obtained from participants after they finished cycling are shown in Table 2.

3.2.1. Attention

The results of a repeated-measures ANOVA indicated that there was a statistically significant main effect of test time on RT ($F_{1,72} = 5.635, p = .020, \eta_p^2 = 0.073$), with a decrease in RT after cycling, and a statistically significant interaction for form perception $\times$ color perception $\times$ test time ($F_{1,72} = 5.586, p = .021, \eta_p^2 = .072$). As shown in Fig. 2, RT was significantly decreased after cycling among participants in the green tree group ($t = 3.334, p = .004$, Cohen's *d* = 0.765, 95% *CI* difference = 8.72–38.44), but not in the other three groups.

3.2.2. Total mood disturbance

The results of a GEE revealed a statistically significant effect of test time on TMD scores (Wald $\chi^2 = 29.084, p < .001$), with a significant decrease after cycling, and a statistically significant interaction between test time and form perception (Wald $\chi^2 = 6.291, p = .012$). Further sub-analyses indicated a significant decrease after cycling with trees ($p < .001$, 95% *CI* difference = $-35.44 \sim -17.04$) and without trees (i.e., “blank”) ($p = .041$, 95% *CI* difference = $-18.79 \sim -0.37$), but no other significant differences in TMD scores after cycling ($p = .108$; Fig. 3).

3.2.3. POMS subscales

The results of a repeated-measures ANOVA revealed a statistically significant main effect of test time on the esteem subscale ($F_{1,72} = 11.727, p = .001, \eta_p^2 = 0.140$), with a significant increase after cycling, and a statistically significant interaction between form perception and

Table 2

Attention, mood states and self-esteem after cycling (post-cycling) ($M \pm SD$).

Test Items	Conditions			
	Blank		Tree	
	Achromatic	Green	Achromatic	Green
Selective Attention (ms)	341.89 $\pm$ 71.15	360.89 $\pm$ 67.00	350.16 $\pm$ 65.61	329.16 $\pm$ 66.96
Mood States (scores)				
Total mood	217.84 $\pm$ 48.85	208.37 $\pm$ 33.28	205.84 $\pm$ 35.94	194.26 $\pm$ 22.58
disturbance	41.05 $\pm$ 7.88	40.74 $\pm$ 7.59	38.53 $\pm$ 5.55	36.95 $\pm$ 3.03
Tension	41.89 $\pm$ 7.30	40.16 $\pm$ 5.64	38.74 $\pm$ 3.66	39.16 $\pm$ 3.58
Anger	44.16 $\pm$ 11.11	42.26 $\pm$ 8.75	43.47 $\pm$ 7.73	40.00 $\pm$ 6.66
Fatigue	43.79 $\pm$ 6.53	42.74 $\pm$ 5.81	42.95 $\pm$ 6.35	41.42 $\pm$ 3.01
Depression	50.74 $\pm$ 12.55	51.68 $\pm$ 6.12	51.37 $\pm$ 9.96	54.32 $\pm$ 7.64
Vigor	45.89 $\pm$ 9.62	43.00 $\pm$ 7.48	44.16 $\pm$ 9.63	42.16 $\pm$ 7.83
Confusion	48.21 $\pm$ 11.82	48.84 $\pm$ 5.55	50.63 $\pm$ 7.76	51.11 $\pm$ 8.79
Esteem	33.58 $\pm$ 3.96	32.21 $\pm$ 3.36	32.95 $\pm$ 4.26	33.12 $\pm$ 3.29
Self-esteem (scores)				

test time ($F_{1,72} = 6.550, p = .013, \eta_p^2 = 0.083$). As shown in Fig. 4, scores for participants on the esteem subscale significantly increased after cycling with trees ($t = -4.234, p < .001$, Cohen's *d* = 0.687, 95% *CI* difference = $-6.46 \sim -2.28$), with no significant differences in any subscale score after cycling without trees (blank).

The results of a GEE revealed a statistically significant effect of test time on the anger subscale (Wald $\chi^2 = 21.469, p < .001$), with a significant decrease after cycling, and a statistically significant interaction between test time and color perception (Wald $\chi^2 = 4.938, p = .026$). Further sub-analyses indicated a significant decrease after cycling with the color green ($p < .001$, 95% *CI* difference = $-5.46 \sim -2.17$), but no significant differences after cycling for the achromatic color (Fig. 5).

The results of the GEEs indicated that test time showed a statistically significant effect on tension (Wald $\chi^2 = 31.256, p < .001$), depression (Wald $\chi^2 = 5.103, p = .024$), and confusion (Wald $\chi^2 = 33.628, p < .001$) subscale scores. Further sub-analyses indicated that scores for the tension ($p < .001$, 95% *CI* difference = $-5.63 \sim -2.71$), depression ($p = .024$, 95% *CI* difference = $-3.02 \sim -0.21$), and confusion ($p < .001$, 95% *CI* difference = $-6.95 \sim -3.44$) subscales were decreased after cycling.

3.2.4. Self-esteem

The results of a GEE revealed a statistically significant effect of test time on scores for the RSES (Wald $\chi^2 = 23.053, p < .001$), with a significant increase after cycling, and a statistically significant interaction

Table 1

Attention, mood states and self-esteem before cycling (pre-cycling) ($M \pm SD$).

Test Items	Conditions				F/χ^2	p
	Blank		Tree			
	Achromatic	Green	Achromatic	Green		
Selective Attention (ms)	365.95 ± 88.63	358.84 ± 76.24	352.32 ± 50.19	352.74 ± 75.44	.142	.934
Mood States (scores)						
Total mood disturbance	220.32 ± 39.82	225.05 ± 34.32	229.00 ± 39.87	223.58 ± 31.55	.505	.918
Tension	42.37 ± 6.70	44.84 ± 8.35	43.37 ± 6.90	43.26 ± 6.47	.759	.859
Anger	41.58 ± 5.78	43.63 ± 7.83	41.74 ± 5.32	43.32 ± 5.74	2.717	.437
Fatigue	42.74 ± 7.50	44.11 ± 7.39	43.68 ± 5.73	43.16 ± 8.28	1.086	.780
Depression	44.21 ± 6.53	44.37 ± 5.92	45.68 ± 7.46	43.11 ± 4.70	.773	.856
Vigor	51.26 ± 10.31	51.68 ± 5.59	50.63 ± 8.64	50.95 ± 8.73	1.056	.788
Confusion	47.95 ± 10.57	48.21 ± 7.15	51.21 ± 11.54	48.63 ± 8.41	1.043	.791
Esteem	47.37 ± 9.47	48.42 ± 5.91	46.05 ± 8.49	46.95 ± 9.24	.260	.854
Self-esteem (scores)	33.11 ± 3.73	31.68 ± 2.79	32.00 ± 4.85	31.47 ± 2.89	.747	.527

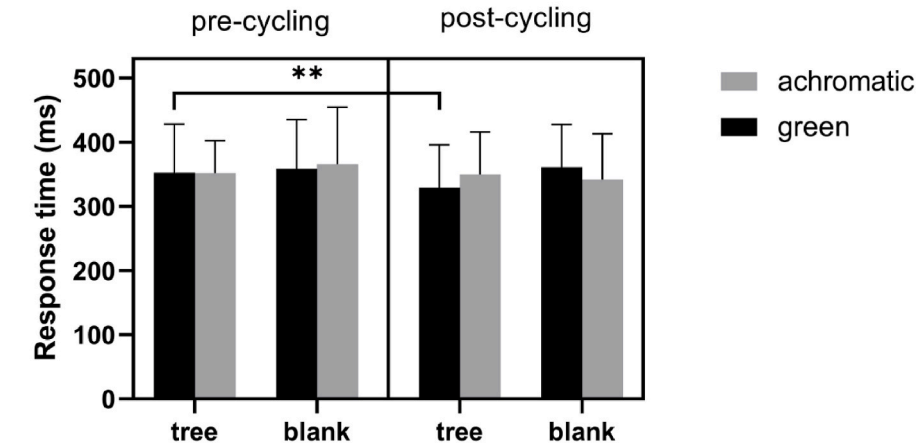


Fig. 2. Mean response times for participants on the attention task by experimental condition. Repeated-measures ANOVA results indicate a significant interaction for test time, form perception, and color perception. $**p < .01$ for the indicated conditions. Error bars indicate one SD.

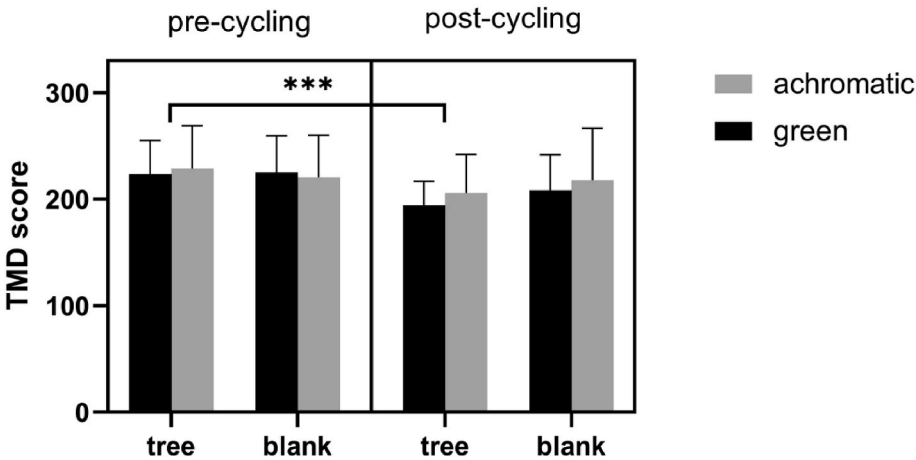


Fig. 3. Profile of Mood States Total Mood Disturbance (TMD) mean scores by experimental condition. The interaction between test time and form perception is significant. $***p < .001$ for the indicated conditions. Error bars indicate one SD.

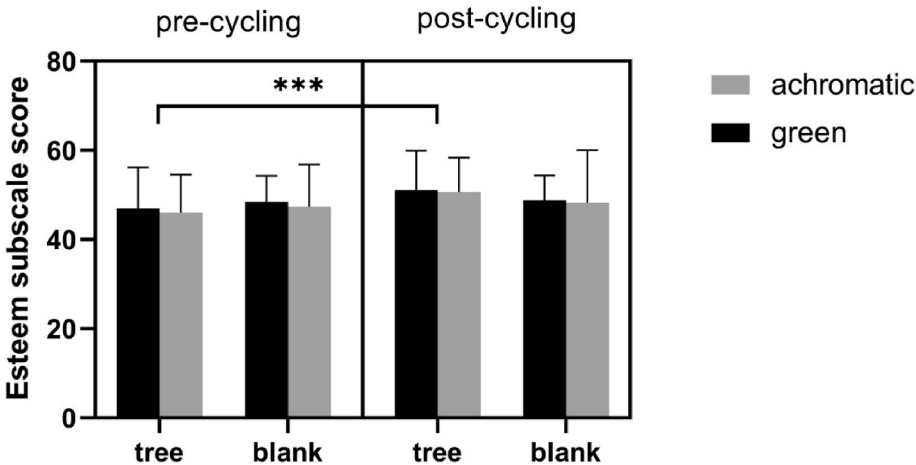


Fig. 4. Differences in mean esteem subscale scores by experimental condition. The interaction between test time and form perception is significant. $***p < .001$ for the indicated conditions. Error bars indicate one SD.

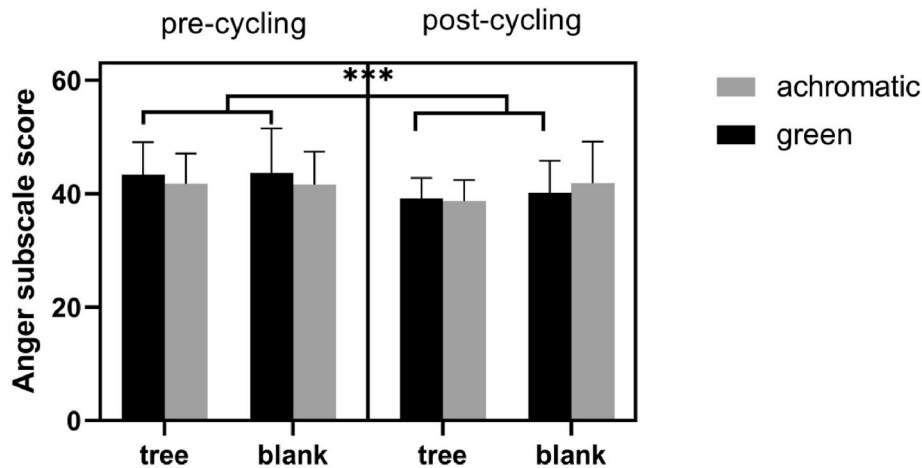


Fig. 5. Differences in anger subscale scores by experimental condition. The interaction between test time and color perception is significant. *** $p < .001$ for the indicated conditions. Error bars indicate one SD.

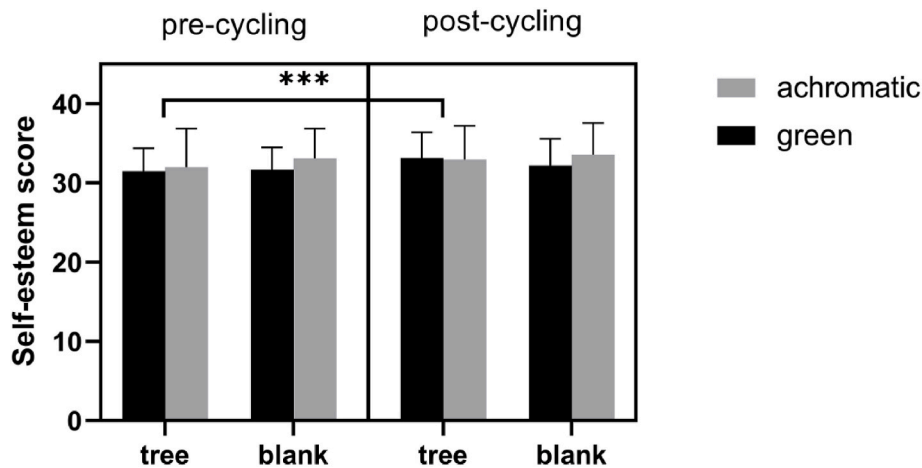


Fig. 6. Difference in mean self-esteem scores by experimental condition. The interaction between test time and form perception is significant. *** $p < .001$ for the indicated conditions. Error bars indicate one SD.

between test time and form perception (Wald $\chi^2 = 4.653$, $p = .031$). Further sub-analyses indicated a significant increase in RSES scores after cycling with trees ($p < .001$, 95% CI difference = 0.85–1.79), but no significant differences in RSES scores after cycling without trees (blank) (Fig. 6).

4. Discussion

Although a large body of evidence has shown that green exercise provides mental health benefits across the life span, to the best of our knowledge, few studies have explored which visual perceptions contribute to the green exercise effects. The present study separated critical visual features, including form (trees with wide canopies and short trunks) and color (green), from the natural environment to determine the impact of form and color perception on the effects of green exercise. The results showed decreased RT on the attention task, lower TMD scores, lower anger, tension, depression, and confusion subscale scores, but higher scores on the esteem subscale and RSES after completing incremental cycling. The RTs on the attention task were significantly decreased after cycling compared with before cycling in the green tree group. Lower TMD and higher scores on the esteem subscale and the RSES were found after cycling with trees, while scores on the anger subscale decreased after cycling with the color green.

4.1. Combination of form and color perception in green exercise

We observed significantly improved attention as assessed in the attention task only for the green tree experimental condition, suggesting that form of trees and color of green are important elements to improve attention restoration. This is consistent with previous experimental studies that have included green trees in both natural and simulated natural environments and have demonstrated beneficial green exercise effects (Olafsdottir et al., 2017; Rogerson & Barton, 2015). Lin, Tsai, Sullivan, Chang, and Chang (2014) conducted an experiment by changing the degree of the participant's awareness of trees to analyze how trees impact attention. The authors found that raising awareness of trees helped to improve attention restoration. The participants in the present study were required to cycle while paying attention to images, which heightened their awareness of the images. The present result was consistent with this previous work, confirming that attention restoration associated with green trees is a valid green exercise effect. Furthermore, our results also revealed that the restorative effect on attention after green exercise did not depend only on the form of trees but also on the color green.

According to ART, the natural environment, beneficial for its soft fascination, is an ideal place to recover from directed attention fatigue. Soft fascination is described by ART as the modest attentional hold necessary to withdraw lingering, unsolved, or unnecessary thoughts and

leave sufficient space for essential mental activity (Basu et al., 2019). In the natural environment, soft fascination enables involuntary attention, which does not require sustained effort or inhibition of competing stimuli and thereby leaves ample opportunity for restoring the capacity of directed attention (Kaplan, 1995). The green tree, which is a familiar element of natural environments, is likely to evoke soft fascination. It loosely captures attention to improve attention performance. Notably, our results showed that a single visual perception, either achromatic trees or the color green, seem to be less restorative than the combined visual perceptions of green trees.

Constructivists assume that the well-being or health promoting effects of nature are acquired through the semantic idea of nature and its restorative abilities. The recognition and association of features with the concept of nature may influence the preference and restoration of a specific natural environment (Van Hedger et al., 2019). Compared with the other conditions used in the present study, the green tree undoubtedly exhibited the rich visual features of a tree, which is a familiar natural element. Rich visual features that are more easily associated with the concept of nature may elicit more positive experiences, conferring a restorative effect. This assumption together with our findings suggests that urban planners and designers should consider designing green spaces with green trees, such as pocket parks, around places that lead to attention fatigue, such as workplaces and schools, to meet the demand for green exercise and thus improve attention restoration.

4.2. Form perception in green exercise

Besides a combination of form and color perception, a single visual perception may also impact green exercise effects. The present results indicated that lower TMD and higher esteem and self-esteem were reported after cycling with images of trees. A previous study explored how a single visual perception contributes to green exercise effects, and the authors believe that the improved mood states are attributed to the color green (Akers et al., 2012). However, their green condition comprised not only the color green but also the form of the tree. Thus, improvements in mood state may not critically depend on the color green. Indeed, subsequent studies failed to find positive effects of perceiving the color green on mood states (Briki & Majed, 2019). The present study examined the role of the color green and the form of tree presented independently. Cycling with a visual stimulus of a tree resulted in lower TMD and higher esteem and self-esteem, suggesting that positive effects on some mood states and self-esteem depend less on the color green and more on the form of trees.

Compared with exposure to inanimate objects, exposure to trees has been shown to improve positive affect and to reduce negative affect (Lohr et al., 1996; Lohr & Pearson-Mims, 2006). The present results suggest that improvements in mood states and self-esteem were likely due to the form of trees. Participants were familiar with viewing achromatic images, such as achromatic photographs and achromatic film, that were sufficient to provide visual features. Exposure even to achromatic trees provides complex form features, including global shape and texture, which often define an object's identity (Taylor & Xu, 2021). According to constructivists, rich visual features are more easily recognized and associated with the concept of nature, which may elicit more positive experiences and thus provide additional exercise benefits to improve mood states and self-esteem.

4.3. Color perception in green exercise

The present study also showed that the color green diminished anger. By contrast, previous studies found no difference in anger between participants who viewed a green color or an achromatic color (white or gray) after physical activity (Akers et al., 2012; Briki & Majed, 2019). However, the finding of the present study is consistent in general with the results of studies examining the response to trees in different canopy

colors. For example, one study found that although viewing a tree canopy in any color was calming, the color green was particularly calming (Kaufman & Lohr, 2004). Green is a positively valenced color and is considered a primitive visual feature of forests in Asian cultures (Saito, 1996) and thus elicits positive affect, such as relaxation, calmness, happiness, comfort, peace, and hope (Kaya & Epps, 2004). In the context of physical activity, studies have also shown that perceiving green increases enjoyment (Briki et al., 2015) and decreases mood disturbance and perceived exertion (Akers et al., 2012). This positive affect may diminish participants' anger.

The exercise benefits of perceiving green are commonly thought to be caused by the connection between the color green and the concept of nature. The findings of the present study provide insights for green exercise. The color green was sufficient to diminish anger, suggesting that the effect of greenness on anger was available independent of a natural element (e.g., trees). However, how people respond to certain colors may vary by individuals' cultural experiences (Chebat & Morrin, 2007). Since the present study was limited to a Chinese cultural context, further investigation and experimentation should be conducted to investigate cross-cultural differences in the green exercise benefits of visual features in the natural environment.

Data from several sources have identified effects of green exercise on attention and mood states. The present study notably provided evidence indicating that a single visual feature (e.g., form or color) was sufficient to improve mood states and self-esteem, whereas combined visual features were required for attention restoration. This finding is consistent with previous studies that detected no correlation between changes in attention tasks and mood states, suggesting that cognitive and affective benefits from nature are due to different processes (Berman et al., 2012; Li & Sullivan, 2016). The findings of the present study further suggested that green exercise influences attention and mood gains from different visual features of the natural environment.

ART asserts that restoration comes from effortless attention to a natural environment, leaving ample opportunity for recovery from directed attention fatigue (Kaplan, 1995). The SRT argues that recovery results from positive changes in emotional state and physiological activity levels that occur in response to the natural environment (Ulrich et al., 1991). The major difference between these two pathways of restoration is that attention restoration is through human cognitive operations, whereas stress recovery is through positive physiological responses to nature. The different cognitive operations or physiological responses associated with the visual features of trees may lead to separable benefits in attention or mood states. The evidence points to visual features that landscape planners, designers, and researchers can manipulate in order to promote exercise benefits. Future research can examine the activity of various brain regions (e.g., ventral frontal cortex, temporal cortex) or physiological responses (e.g., cortisol level, skin conductance level) to different visual features of the natural environment from a cognitive neuroscience perspective to understand the mechanisms of those behavioral performances.

This study had some potential limitations. First, although the heart rate and perceived exertion were employed to monitor exercise intensity, they were not included further in data analysis. Further studies which take exercise intensity as a variable might investigate whether the visual perceptions affect perceived and objective exercise intensity. Furthermore, research that includes different exercise programs, such as exercise at individual pace or different duration, could be valuable to supplement the findings of this study. Second, this study manipulated form perception by the presence or absence of tree form, and preliminary results provide direct evidence that the form of trees facilitates more exercise benefits, especially for mood and self-esteem. There is abundant room for further progress in investigating whether the form of trees provides benefits for green exercise compared with other forms, such as inanimate objects or other plants. In spite of its limitations, however, the study certainly adds to our understanding of how visual perception—form and color—contribute to the effects of green exercise.

5. Conclusion

The present study is, to our knowledge, the first to show that the effects of green exercise on mood and attention grew from single or combined visual perceptions. That is, perception of the tree form improved total mood disturbance and self-esteem, whereas perception of the color green diminished anger, and the perception of form and color together improved attention. Our findings may inform urban green space planning to maximize the benefits of green exercise.

CRedit authorship contribution statement

Danxuan Zhang: research design, conceptualization, methods, data collection and analysis, writing original draft and modification. **Xin-hong Jin:** conceptualization, methods, and draft modification. **Luning Wang:** conceptualization, data collection, and draft modification. **Yahong Jin:** research design, conceptualization, methods, research administration, and draft modification.

Ethics statement

This study was approved by the Institutional Review Board at Tianjin University of Sport, and was conducted in accordance with the Declaration of Helsinki. Prior to any data collection, all participants were thoroughly informed about the study and provided written informed consent.

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Declaration of interest statement

The authors have no conflict of interest to declare.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvp.2023.102028>.

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